

GED Science Practice Test 9

PART A — QUESTIONS

GED-SCI9-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student investigates how oxygen availability affects yeast cells. Two sealed flasks contain the same amount of yeast and glucose solution. Flask A has oxygen pumped into it before sealing. Flask B has no added oxygen. After 20 minutes, the student measures the amount of ethanol produced.

Flask | Oxygen added? | Ethanol produced (mL)

A | Yes | 1

B | No | 9

Which conclusion is best supported by the data?

- A) Yeast produced more ethanol when oxygen was limited.
- B) Yeast produced more ethanol when oxygen was added.
- C) Oxygen level had no relationship to ethanol production.
- D) Yeast cannot use glucose in any condition.

GED-SCI9-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the speed of a cart at different times.

Time (s): 0, 2, 4, 6

Speed (m/s): 1, 5, 9, 13

What is the cart's average acceleration from 0 to 6 seconds?

- A) 1 m/s^2
- B) 2 m/s^2
- C) 6 m/s^2
- D) 12 m/s^2

GED-SCI9-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A scientist studies two nearby areas. Area A is covered by natural grassland. Area B has been

converted into bare, compacted soil by repeated vehicle traffic. During heavy rain, Area B has more surface runoff and less water entering the soil than Area A.

Which explanation best fits the observations?

- A) Compacted soil has fewer connected pore spaces, so less water infiltrates.
- B) Compacted soil creates new rain during storms.
- C) Grassland prevents gravity from acting on water.
- D) Bare soil always absorbs more water than soil with plants.

GED-SCI9-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

During intense exercise, muscle cells may need energy faster than oxygen can be delivered. The cells can temporarily use fermentation to release some energy from glucose. This process produces lactic acid, which can build up in muscles.

Which statement best describes fermentation in muscle cells?

- A) It can release some energy when oxygen supply is limited.
- B) It stores sunlight in chloroplasts.
- C) It removes all glucose from the body permanently.
- D) It happens only in bones and teeth.

GED-SCI9-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A motorbike increases its speed from 4 m/s to 22 m/s in 6 seconds.

Acceleration = change in speed $\div$ time

What is the motorbike's acceleration in m/s²?

Type your answer as a whole number.

GED-SCI9-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how ramp angle affects the final speed of a ball. The student uses the same ball and releases it from the same starting point on ramps set at different angles.

Which TWO factors should be kept constant?

- A) Type of ball

- B) Starting point
- C) Ramp angle
- D) Final speed
- E) Measured result

GED-SCI9-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person with Down syndrome usually has three copies of chromosome 21 instead of two. This condition results from a change in [D1], not from a change in the sequence of a single gene. The extra chromosome can be identified with [D2].

Dropdowns:

D1) chromosome number | soil moisture | energy transfer

D2) a karyotype | a weather map | a pH indicator

GED-SCI9-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A spoon appears bent when placed partly in a glass of water. Light travels at different speeds in air and water, so its direction changes when it passes from one material into the other.

Which wave behavior explains the apparent bending?

- A) Refraction
- B) Combustion
- C) Magnetism
- D) Precipitation

GED-SCI9-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A text model describes Earth's layers. The crust is the thin outer layer. Below it, the mantle contains hot rock that can slowly flow over long periods. The outer core is liquid metal, and the inner core is solid metal under very high pressure.

Which layer is described as liquid metal?

- A) Outer core
- B) Inner core
- C) Crust
- D) Upper atmosphere

GED-SCI9-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A researcher records the number of people reporting pollen allergy symptoms during four weeks.

Week | Average pollen level | People reporting symptoms

1 | Low | 12

2 | Moderate | 31

3 | High | 58

4 | Very high | 74

Which conclusion is best supported by the data?

- A) Reports of allergy symptoms increased as pollen level increased.
- B) Reports of allergy symptoms decreased as pollen level increased.
- C) Pollen level had no relationship to symptoms.
- D) The highest symptom count occurred when pollen level was low.

GED-SCI9-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

- A) Type of lens used
- B) Distance at which the image is focused
- C) Same object placed in front of each lens
- D) Brand of notebook used for recording

Targets:

- T1) Independent variable
- T2) Dependent variable
- T3) Controlled variable

GED-SCI9-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The immune system responds to a small cut when damaged cells release chemical signals. White blood cells move toward the area, and blood vessels nearby may widen. The area can become red, warm, and swollen as the body begins repair and defense.

Which response is mainly described?

- A) Inflammation
- B) Fertilization
- C) Photosynthesis
- D) Fossilization

GED-SCI9-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A sealed balloon is placed in a warm room. The gas particles inside the balloon move faster and collide with the balloon walls more often and more forcefully. The balloon expands slightly.

Which statement best explains the change?

- A) Heating the gas increased particle motion and pressure, causing expansion.
- B) Heating the gas removed all particles from the balloon.
- C) Heating the gas turned it into solid metal.
- D) Heating the gas stopped all collisions.

GED-SCI9-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A satellite completes 12 orbits around Earth in 36 hours.

What is the average time for one orbit in hours?

Type your answer as a whole number.

GED-SCI9-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether a certain soil fungus affects tomato root growth. One group of tomato plants is grown in soil with the fungus. A second group is grown in the same type of soil without the fungus. All plants receive the same light, water, and pot size. After four weeks, root length is measured.

Why is the group without the fungus included?

- A) It provides a comparison for the fungus-treated group.
- B) It guarantees that roots will not grow.
- C) It changes the fungus into plant tissue.
- D) It eliminates the need to measure root length.

GED-SCI9-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student compares how quickly different metals conduct heat. Rods of copper, steel, and brass are

placed with one end in hot water. A small wax bead is attached to the other end of each rod. The student records the time until each wax bead melts.

What is the dependent variable?

- A) Time until the wax bead melts
- B) Type of metal rod
- C) Temperature of the hot water
- D) Size of the wax bead

GED-SCI9-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

When two magnets are brought near each other, opposite poles [D1], while like poles [D2].

Dropdowns:

D1) attract | evaporate | dissolve

D2) repel | combine into water | disappear

GED-SCI9-018

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A lab technician tracks oxygen use by cells at different temperatures.

Temperature (°C): 15, 25, 35, 45

Oxygen used in 10 min (mL): 6, 14, 29, 13

Which temperature was associated with the highest oxygen use?

- A) 15°C
- B) 25°C
- C) 35°C
- D) 45°C

GED-SCI9-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the input force needed to lift the same load using different pulley setups.

Pulley setup | Input force needed (N)

- A | 120
- B | 80
- C | 60
- D | 40

Which setup gives the greatest mechanical advantage?

- A) Setup A
- B) Setup B
- C) Setup C
- D) Setup D

GED-SCI9-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A city compares two transportation patterns and carbon dioxide emissions.

Year	Public transit trips (millions)	Transportation CO ₂ emissions (thousand tons)
1	24	910
2	31	850
3	39	790
4	46	730

Which TWO statements are best supported by the data?

- A) Public transit trips increased from Year 1 to Year 4.
- B) Transportation CO₂ emissions decreased from Year 1 to Year 4.
- C) CO₂ emissions increased as public transit trips increased.
- D) Public transit trips stayed the same every year.
- E) Year 4 had the highest transportation CO₂ emissions.

GED-SCI9-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person has a damaged heart valve that does not close completely. Some blood leaks backward each time the heart pumps. As a result, less blood may move forward to deliver oxygen and nutrients to body tissues.

Which function is most directly affected?

- A) Circulation of blood through the body
- B) Breakdown of starch in the mouth
- C) Exchange of genetic information during meiosis
- D) Formation of sedimentary rock layers

GED-SCI9-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student observes water boiling in an open pot. The temperature of the water remains near 100°C

while boiling continues. Added energy is used to change liquid water into water vapor instead of raising the temperature.

Which statement best explains the constant temperature during boiling?

- A) Added energy is used for a phase change.
- B) Added energy disappears and is not transferred.
- C) The water stops containing particles.
- D) The water becomes colder while boiling.

GED-SCI9-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A volcano releases ash and sulfur dioxide high into the atmosphere during a large eruption. Some particles and gases can reflect sunlight back to space for a short time, reducing the amount of solar energy reaching Earth's surface.

Which short-term effect could result from a large eruption?

- A) Temporary cooling of surface temperatures
- B) Permanent stopping of Earth's rotation
- C) Elimination of gravity
- D) Immediate formation of all fossil fuels

GED-SCI9-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A survey of 320 students finds that 88 students report having asthma.

What percent of the students report having asthma?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI9-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student wants to test how blade angle affects the lift produced by a small model propeller. The student changes the blade angle but keeps the motor speed and blade length the same. The lift force is measured for each angle.

Which factor is the independent variable?

- A) Blade angle
- B) Lift force

- C) Motor speed
- D) Blade length

GED-SCI9-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A plant cell has a large central vacuole filled with water and dissolved substances. When the vacuole is full, it presses against the cell wall and helps keep the plant upright.

Which statement best describes one function of the central vacuole?

- A) It helps maintain pressure that supports the plant cell.
- B) It pumps blood through plant veins.
- C) It controls the orbit of planets.
- D) It breaks rocks into sediment.

GED-SCI9-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures electrical energy used by four devices in the same amount of time.

Device	Voltage (V)	Current (A)
A	12	1
B	12	2
C	12	3
D	12	4

Power = voltage $\times$ current

Which device has the greatest power?

- A) Device A
- B) Device B
- C) Device C
- D) Device D

GED-SCI9-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records the percentage of land covered by forest in a region.

Year	Forest cover (%)
1990	64
2000	58

2010 | 51

2020 | 43

What was the change in forest cover from 1990 to 2020?

- A) Decrease of 21 percentage points
- B) Increase of 21 percentage points
- C) Decrease of 43 percentage points
- D) Increase of 64 percentage points

GED-SCI9-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A lab compares the average time needed for blood to clot in four samples.

Sample | Clotting time (seconds)

- A | 42
- B | 65
- C | 38
- D | 51

Which sample clotted fastest?

- A) Sample A
- B) Sample B
- C) Sample C
- D) Sample D

GED-SCI9-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student places a copper wire in a blue solution containing silver ions. After some time, silver-colored crystals form on the wire and the solution changes color. Atoms and ions have rearranged to form new substances.

Which type of change occurred?

- A) Chemical change
- B) Physical change only
- C) Change in shape only
- D) Change in position only

GED-SCI9-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A researcher studies whether different salt levels affect bean seedling root growth. Seeds are placed in solutions with low, medium, or high salt levels. The researcher measures root length after five days.

Which TWO steps would improve the investigation?

- A) Use the same type of bean seed in all groups.
- B) Use the same number of seeds at each salt level.
- C) Change light level and salt level at the same time.
- D) Measure only the seedling with the longest root.
- E) Use a different growth time for each group.

GED-SCI9-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the distance traveled by a sound pulse in a material.

Time (s): 1, 2, 3, 4

Distance (m): 1,500, 3,000, 4,500, 6,000

What is the speed of the sound pulse in this material?

- A) 750 m/s
- B) 1,500 m/s
- C) 3,000 m/s
- D) 6,000 m/s

GED-SCI9-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

In a pond, leeches attach to fish and feed on their blood. The leeches benefit by getting food, while the fish may lose energy and become more likely to develop infections.

Which type of relationship is described?

- A) Parasitism
- B) Mutualism
- C) Commensalism
- D) Cooperation between two producers

GED-SCI9-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Some antibiotics interfere with bacterial ribosomes. Human cells have ribosomes too, but bacterial

ribosomes are different enough that certain antibiotics can affect bacteria more strongly than human cells.

What is the main reason these antibiotics can target bacteria?

- A) Bacterial ribosomes differ from human ribosomes.
- B) Bacteria do not contain any genetic material.
- C) Human cells do not make proteins.
- D) Antibiotics turn bacteria into viruses.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI9-001 — Correct: A — Correct Answer: Yeast produced more ethanol when oxygen was limited. — Explanation: Flask B had no added oxygen and produced 9 mL of ethanol, more than Flask A.

GED-SCI9-002 — Correct: B — Correct Answer: 2 m/s^2 — Explanation: Speed increased from 1 m/s to 13 m/s, a change of 12 m/s over 6 seconds. Acceleration = $12 \div 6 = 2 \text{ m/s}^2$.

GED-SCI9-003 — Correct: A — Correct Answer: Compacted soil has fewer connected pore spaces, so less water infiltrates. — Explanation: Compaction reduces pore space and permeability, increasing runoff and reducing infiltration.

GED-SCI9-004 — Correct: A — Correct Answer: It can release some energy when oxygen supply is limited. — Explanation: Fermentation allows cells to release some energy from glucose when oxygen is limited.

GED-SCI9-005 — Correct: 3 — Tolerance: exact — Correct Answer: 3 — Explanation: Change in speed = $22 - 4 = 18 \text{ m/s}$. Acceleration = $18 \div 6 = 3 \text{ m/s}^2$.

GED-SCI9-006 — Correct: A,B — Correct Answer: Type of ball | Starting point — Explanation: These factors should stay the same so ramp angle is the main variable being tested.

GED-SCI9-007 — Correct: D1=chromosome number; D2=a karyotype — Correct Answer: D1=chromosome number; D2=a karyotype — Explanation: Down syndrome usually involves an extra chromosome 21, which can be identified by a karyotype.

GED-SCI9-008 — Correct: A — Correct Answer: Refraction — Explanation: Refraction occurs when light changes direction as it passes between materials.

GED-SCI9-009 — Correct: A — Correct Answer: Outer core — Explanation: Earth's outer core is described as liquid metal.

GED-SCI9-010 — Correct: A — Correct Answer: Reports of allergy symptoms increased as pollen level increased. — Explanation: The number of people reporting symptoms rose from 12 to 74 as pollen level increased.

GED-SCI9-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Type of lens used; T2=Distance at which the image is focused; T3=Same object placed in front of each lens — Explanation: The lens type is changed, focus distance is measured, and the object is controlled.

GED-SCI9-012 — Correct: A — Correct Answer: Inflammation — Explanation: Redness, warmth, swelling, and white blood cell movement are signs of inflammation.

GED-SCI9-013 — Correct: A — Correct Answer: Heating the gas increased particle motion and pressure, causing expansion. — Explanation: Warmer gas particles move faster and collide more forcefully with the balloon walls.

GED-SCI9-014 — Correct: 3 — Tolerance: exact — Correct Answer: 3 — Explanation: $36 \text{ hours} \div 12 \text{ orbits} = 3 \text{ hours per orbit}$.

GED-SCI9-015 — Correct: A — Correct Answer: It provides a comparison for the fungus-treated group. — Explanation: The group without fungus acts as a comparison group to judge the effect of the fungus.

GED-SCI9-016 — Correct: A — Correct Answer: Time until the wax bead melts — Explanation: The dependent variable is the measured outcome after changing the type of metal.

GED-SCI9-017 — Correct: D1=attract; D2=repel — Correct Answer: D1=attract; D2=repel — Explanation: Opposite magnetic poles attract, while like magnetic poles repel.

GED-SCI9-018 — Correct: C — Correct Answer: 35°C — Explanation: The highest oxygen use, 29 mL, occurred at 35°C.

GED-SCI9-019 — Correct: D — Correct Answer: Setup D — Explanation: For the same load, the setup requiring the least input force has the greatest mechanical advantage.

GED-SCI9-020 — Correct: A,B — Correct Answer: Public transit trips increased from Year 1 to Year 4. | Transportation CO₂ emissions decreased from Year 1 to Year 4. — Explanation: Transit trips rose from 24 million to 46 million, while emissions decreased from 910 to 730 thousand tons.

GED-SCI9-021 — Correct: A — Correct Answer: Circulation of blood through the body — Explanation: A damaged valve can reduce the heart's ability to move blood forward efficiently.

GED-SCI9-022 — Correct: A — Correct Answer: Added energy is used for a phase change. — Explanation: During boiling, added energy changes liquid water into vapor instead of increasing temperature.

GED-SCI9-023 — Correct: A — Correct Answer: Temporary cooling of surface temperatures — Explanation: Ash and sulfur compounds can reflect sunlight and temporarily reduce surface heating.

GED-SCI9-024 — Correct: 28 — Tolerance: exact — Correct Answer: 28 — Explanation: $88 \div 320 \times 100 = 27.5\%$, which rounds to 28%.

GED-SCI9-025 — Correct: A — Correct Answer: Blade angle — Explanation: The independent variable is the factor intentionally changed by the student.

GED-SCI9-026 — Correct: A — Correct Answer: It helps maintain pressure that supports the plant cell. — Explanation: A full central vacuole creates pressure against the cell wall, helping support the plant cell.

GED-SCI9-027 — Correct: D — Correct Answer: Device D — Explanation: Power = voltage $\times$ current. Device D has $12\text{ V} \times 4\text{ A} = 48\text{ W}$, the greatest value.

GED-SCI9-028 — Correct: A — Correct Answer: Decrease of 21 percentage points — Explanation: Forest cover decreased from 64% to 43%, a change of 21 percentage points.

GED-SCI9-029 — Correct: C — Correct Answer: Sample C — Explanation: The fastest clotting sample has the shortest clotting time, 38 seconds.

GED-SCI9-030 — Correct: A — Correct Answer: Chemical change — Explanation: New silver-colored crystals and solution color change indicate new substances formed.

GED-SCI9-031 — Correct: A,B — Correct Answer: Use the same type of bean seed in all groups. | Use the same number of seeds at each salt level. — Explanation: Keeping seed type and sample size constant improves the investigation.

GED-SCI9-032 — Correct: B — Correct Answer: 1,500 m/s — Explanation: Speed = distance $\div$ time. The sound pulse travels 1,500 meters each second.

GED-SCI9-033 — Correct: A — Correct Answer: Parasitism — Explanation: The leech benefits while the fish is harmed, which is parasitism.

GED-SCI9-034 — Correct: A — Correct Answer: Bacterial ribosomes differ from human ribosomes. — Explanation: Some antibiotics affect bacterial ribosomes because they are different from human ribosomes.